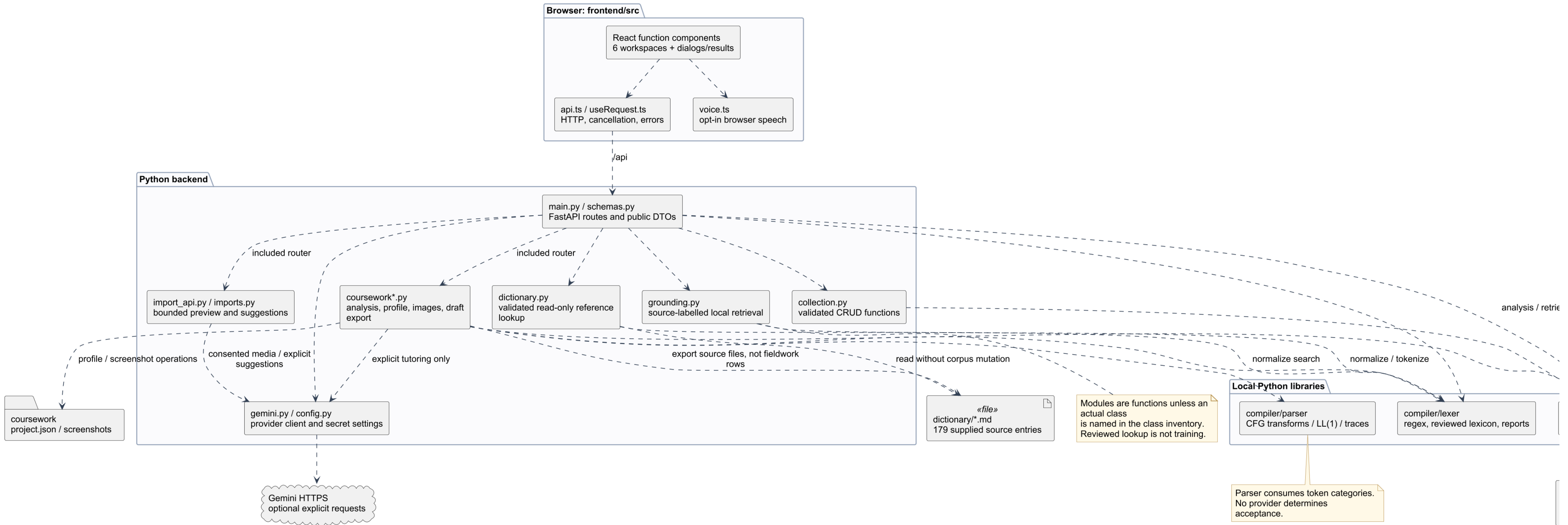
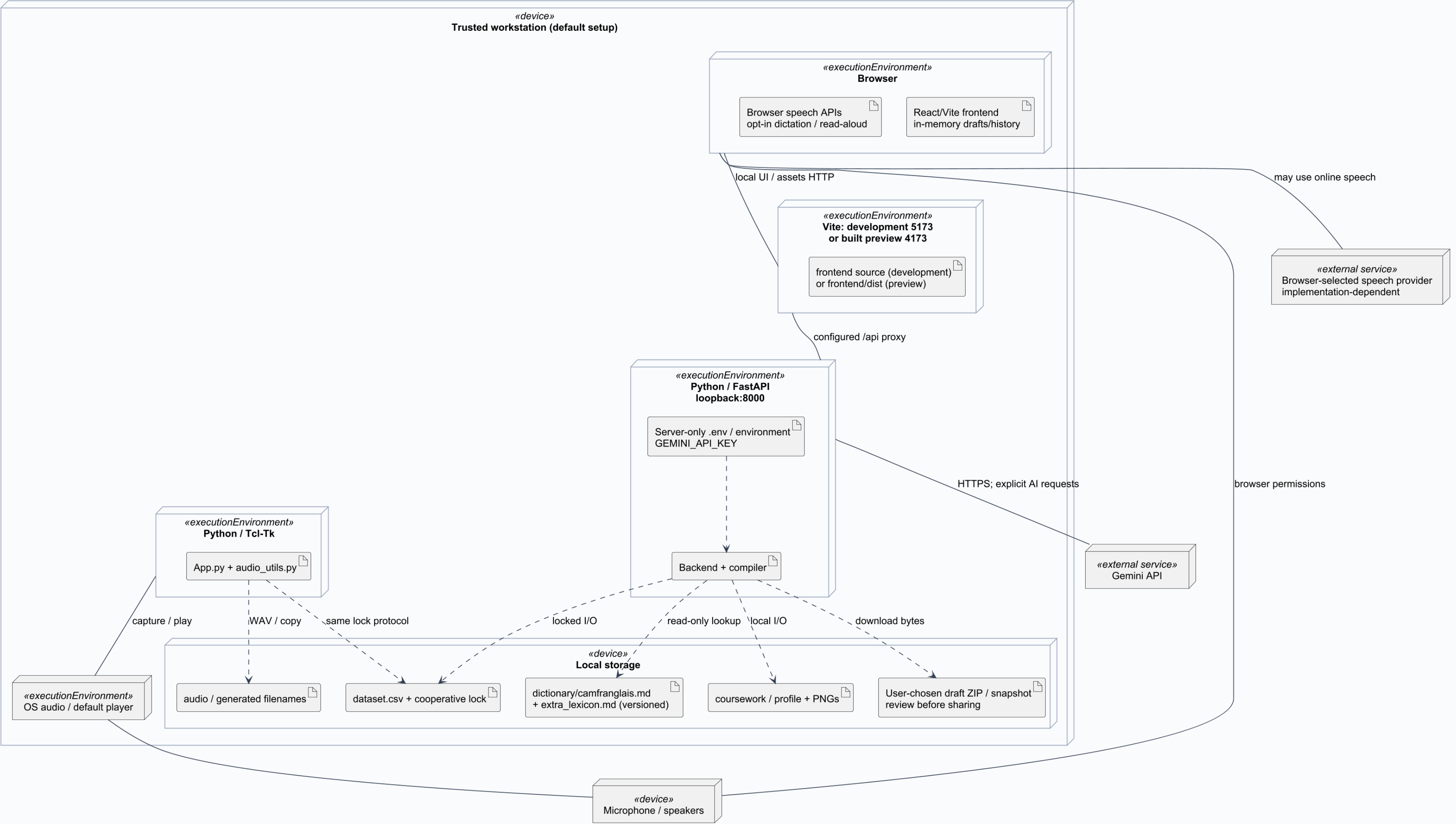


Implemented package dependencies and data boundaries



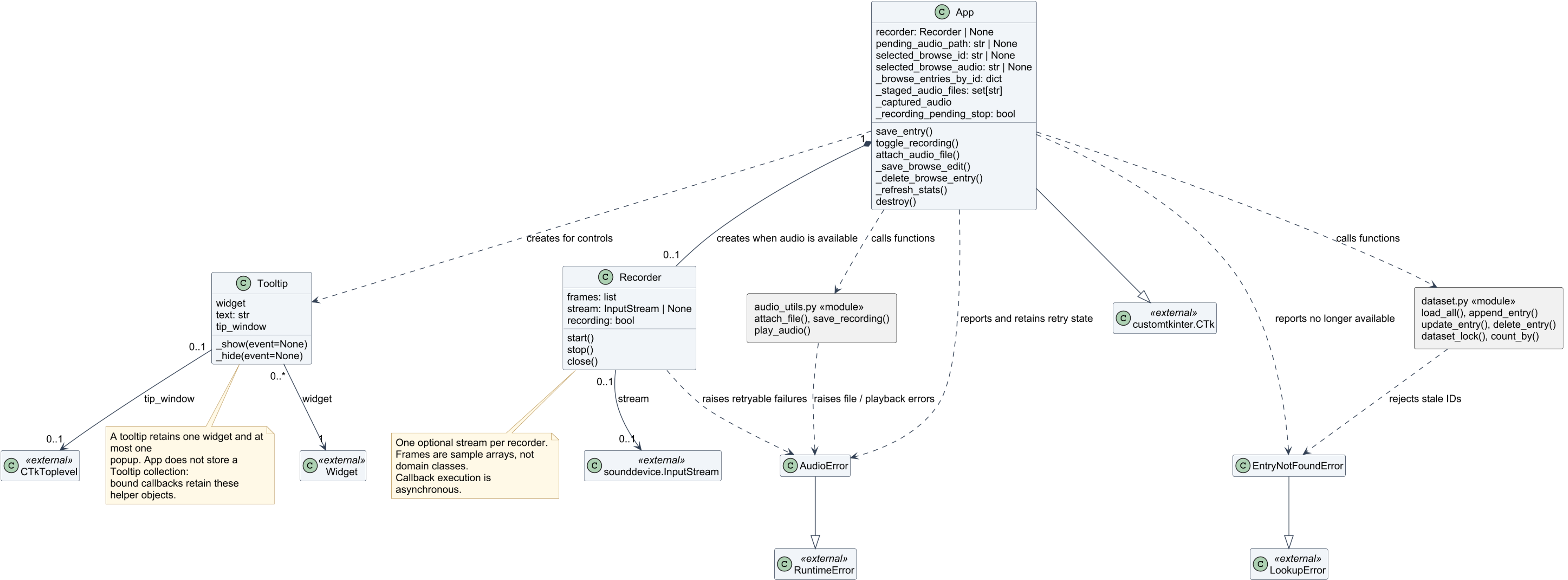
Dashed arrows are dependencies, not timed calls.
Browser speech may contact a separate browser provider (deployment view).
Imports do not automatically write CSV. Export reads saved evidence.

Trusted local deployment; explicit optional remote services



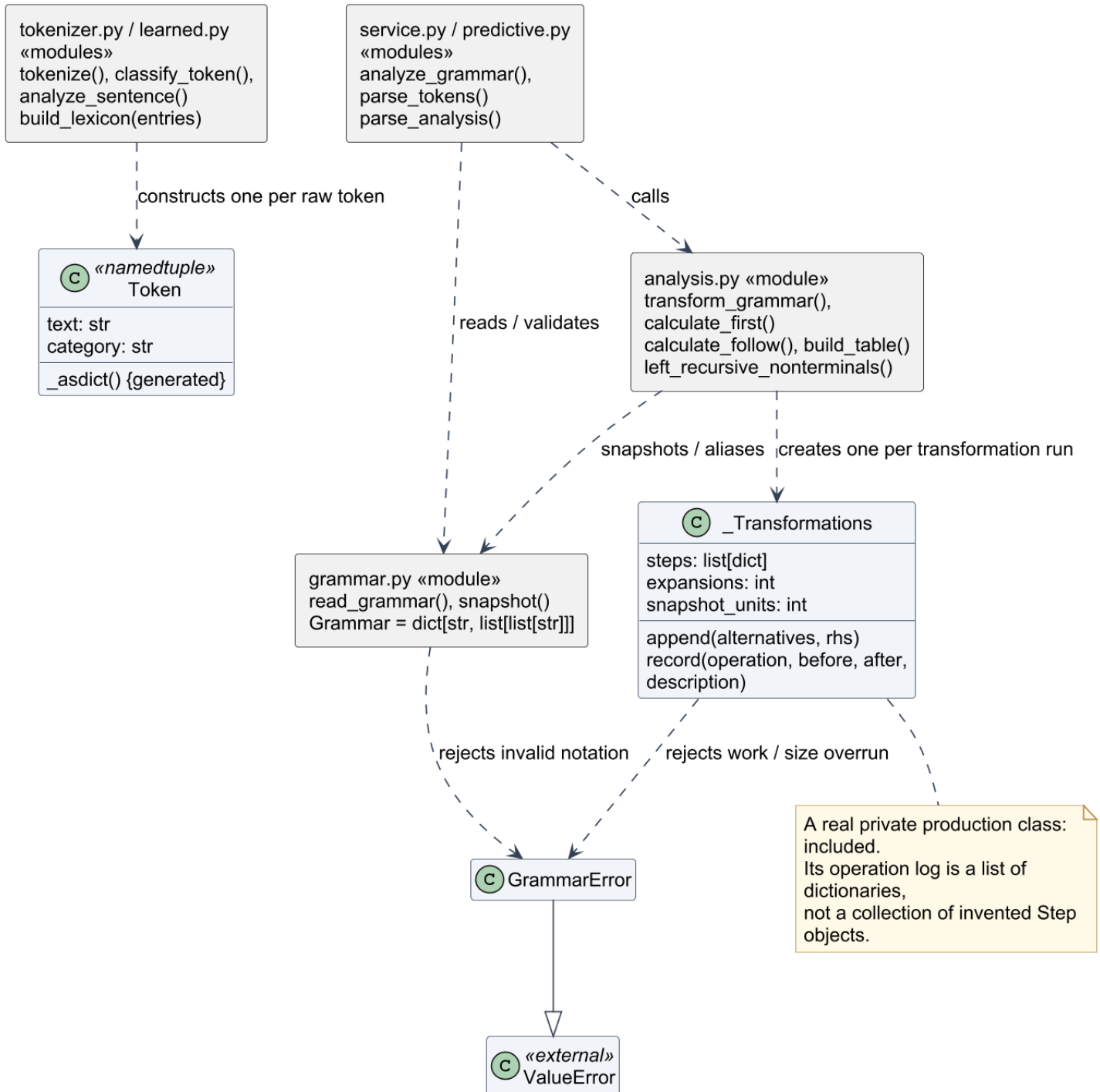
No authentication or public multi-user deployment is supplied.
CORS is not access control. Keep the backend on a trusted local boundary.
If the backend is remote, its CSV/audio/profile live on that server.
FastAPI exposes the API only; Vite serves development or built-preview UI.
Production hosting needs a separate static server and matching API routing.
AI/provider access is optional; local compiler, CRUD and exact reviewed lookup work offline.

Desktop application classes and audio ownership



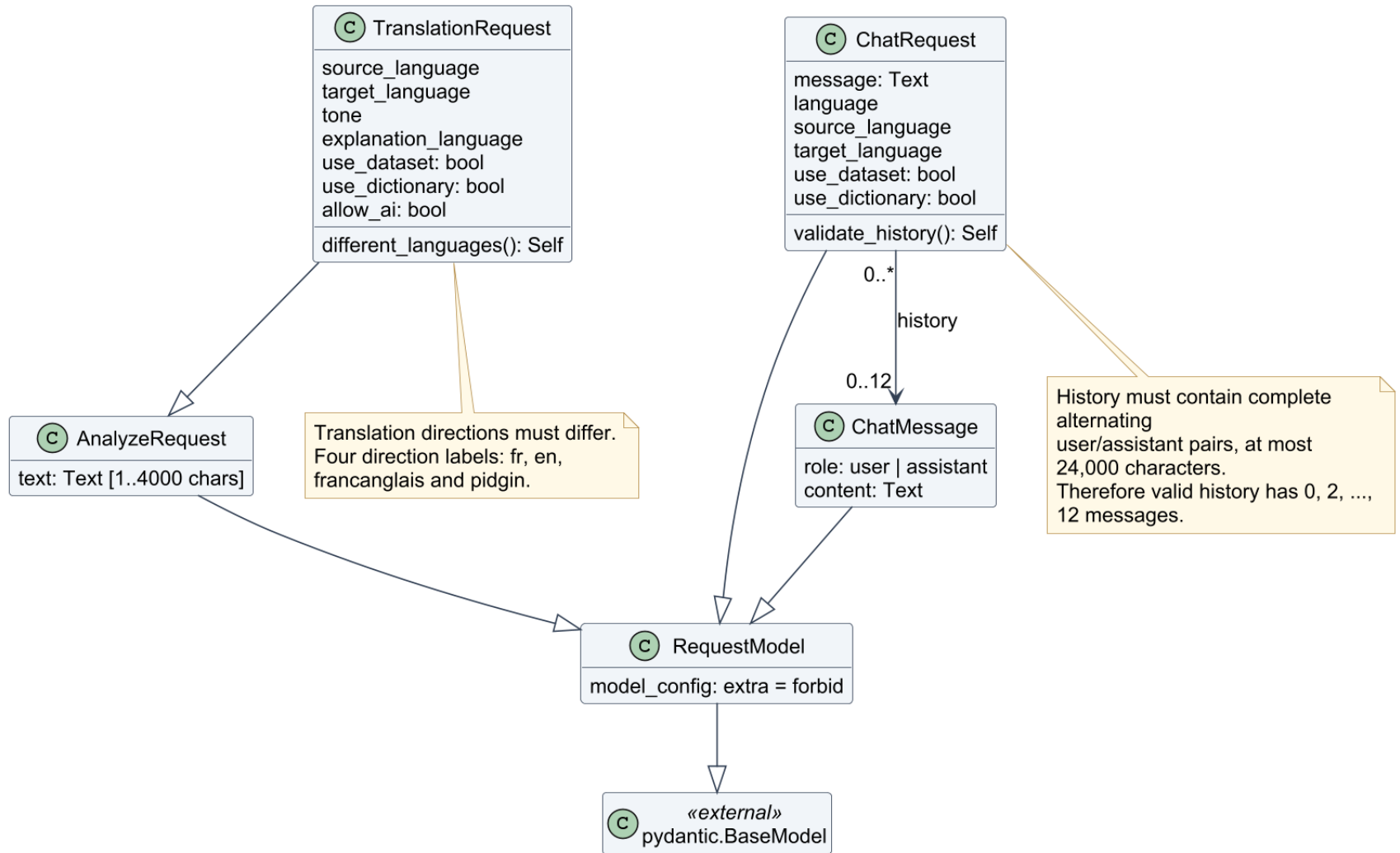
Five production classes; library types are explicitly external.
Per-instance target multiplicities express stored references.
Audio filenames are not object ownership or database foreign keys.

Compiler runtime classes and function modules



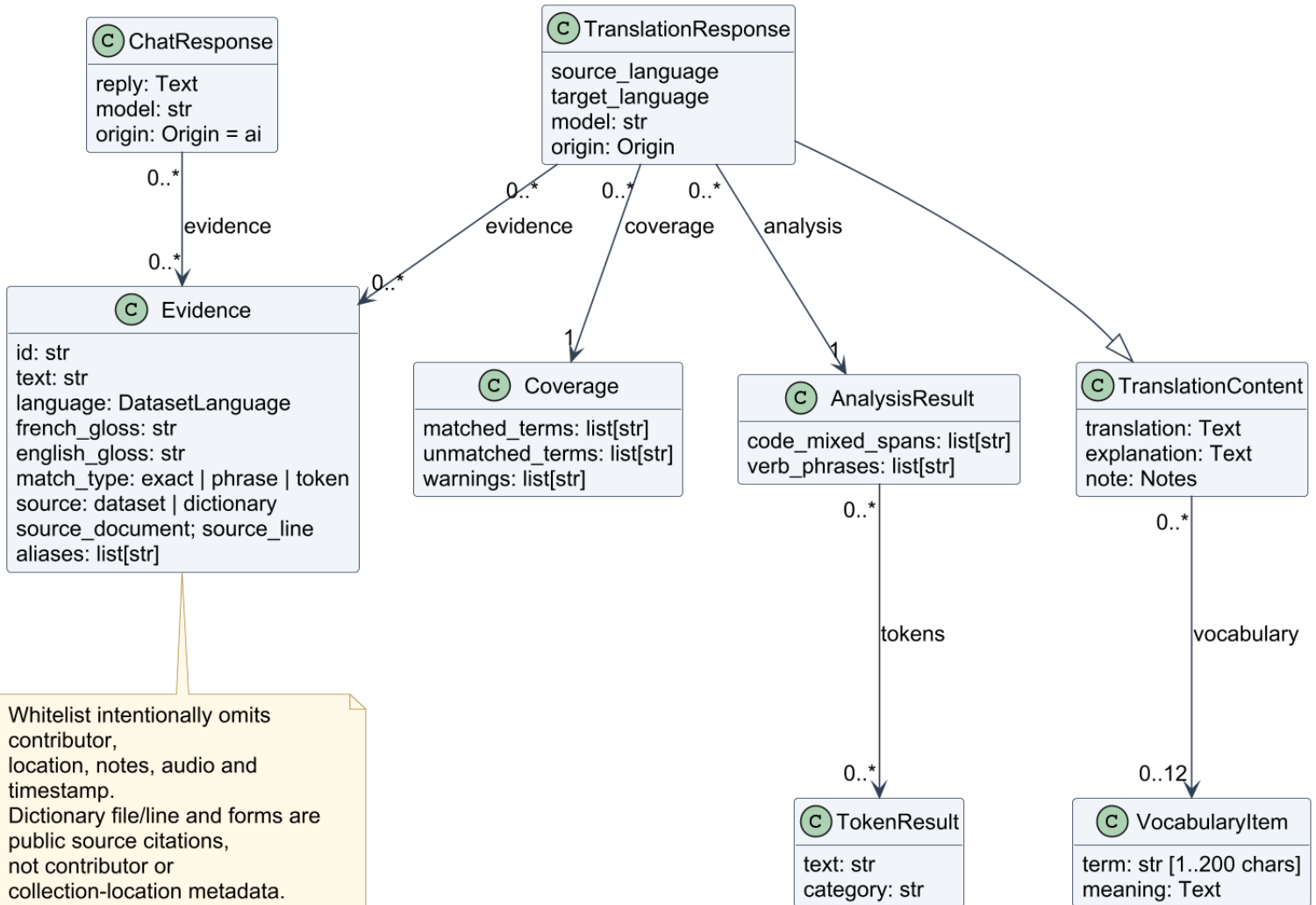
Token is an application-defined runtime class generated by namedtuple. Grammar, Production, analysis results and parse traces are aliases/dictionaries. There is no `LexerService`, `Grammar` class, `Parser` class or compiler database. Lifetimes of function-local values are not drawn as class compositions.

API request inheritance and conversation validation



All five primary classes live in backend.schemas.
Text is a constrained string alias, not an additional class.
Entry, import and coursework request subclasses are in detail views.

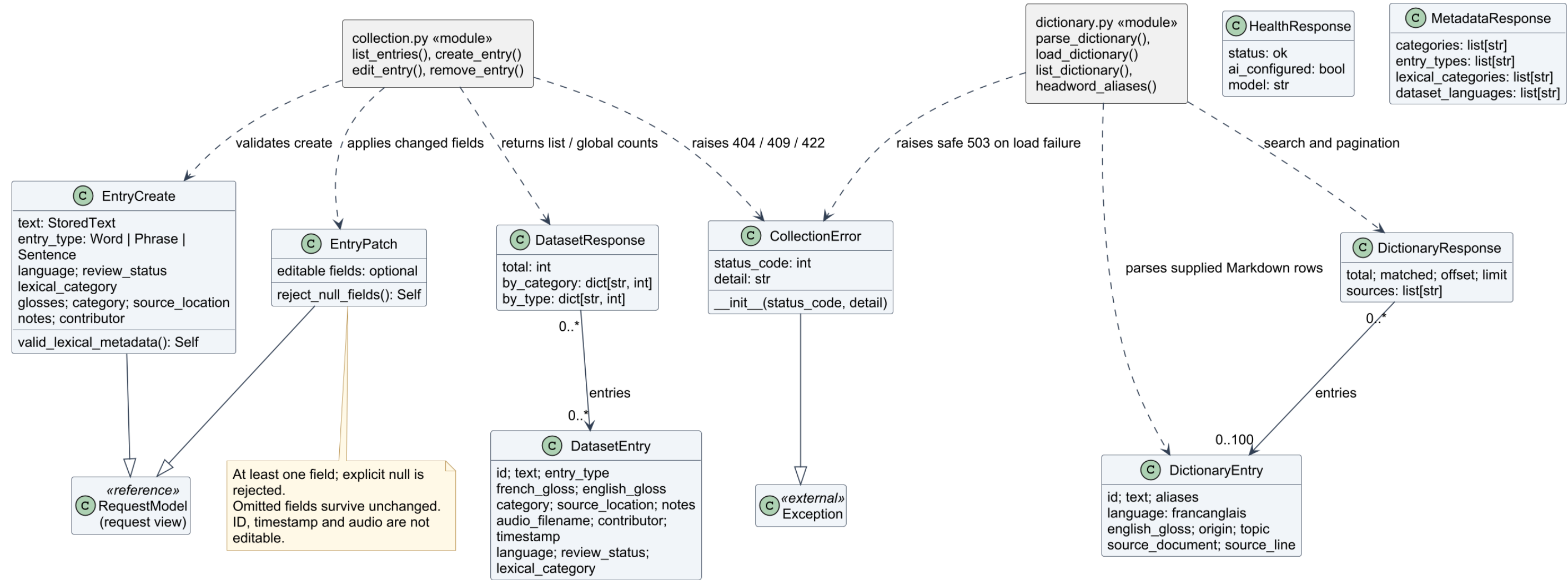
Translation and analysis result DTOs



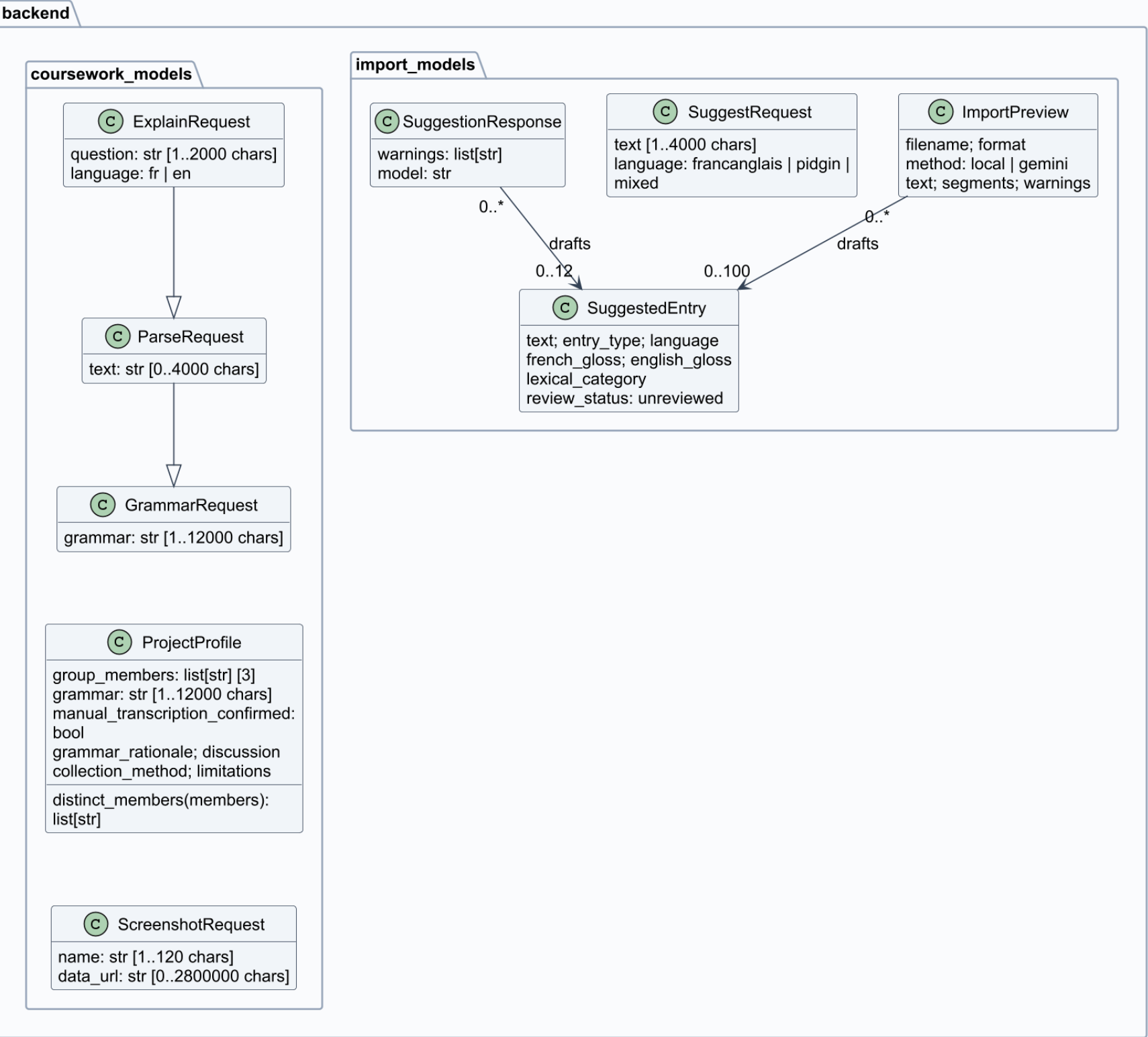
Whitelist intentionally omits contributor, location, notes, audio and timestamp. Dictionary file/line and forms are public source citations, not contributor or collection-location metadata. Retrieval emits at most 8 evidence records, but these DTO list types alone do not enforce 8.

All eight primary classes live in backend.schemas. Each directly extends pydantic.BaseModel except TranslationResponse. Associations describe nested values, not exclusive lifetimes; Pydantic values may be shared/reused by Python callers. Validation/serialization use inherited model_validate/model_dump.

Separate collection and reference dictionary DTOs

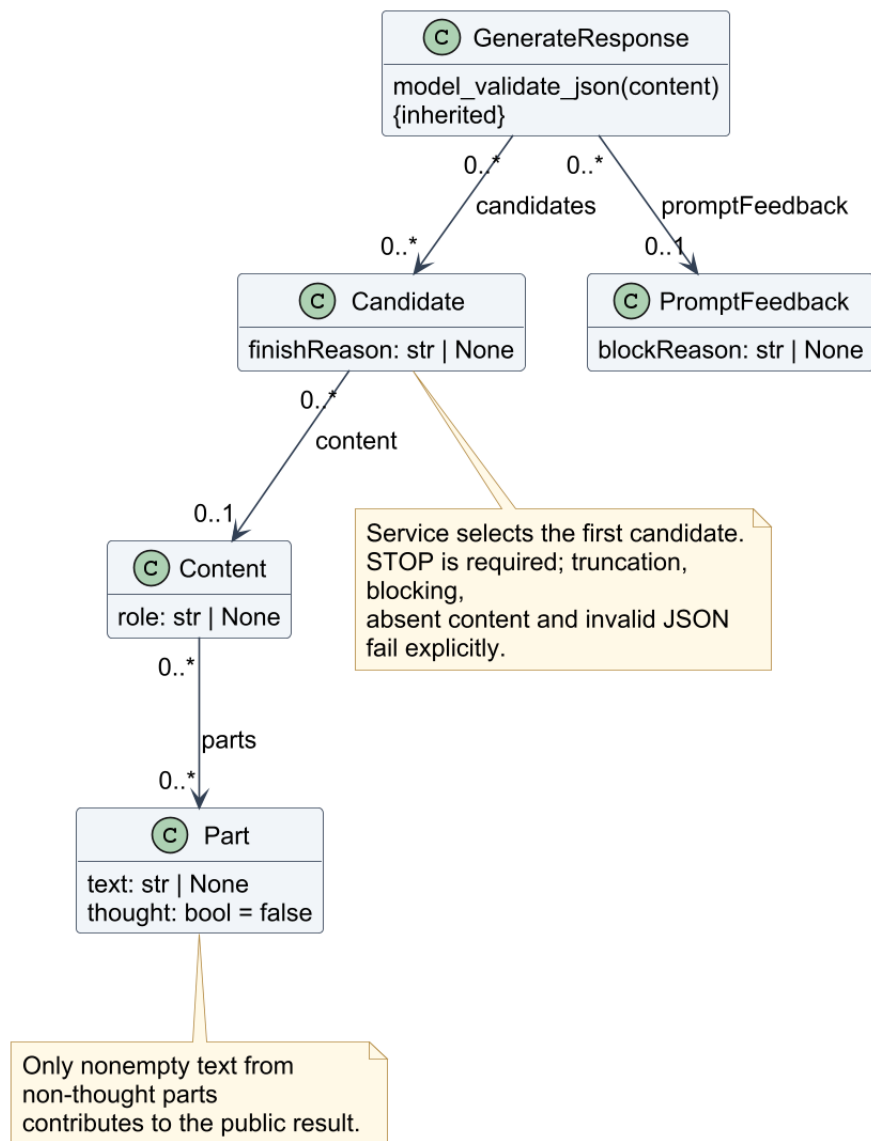


Import and coursework request/result classes



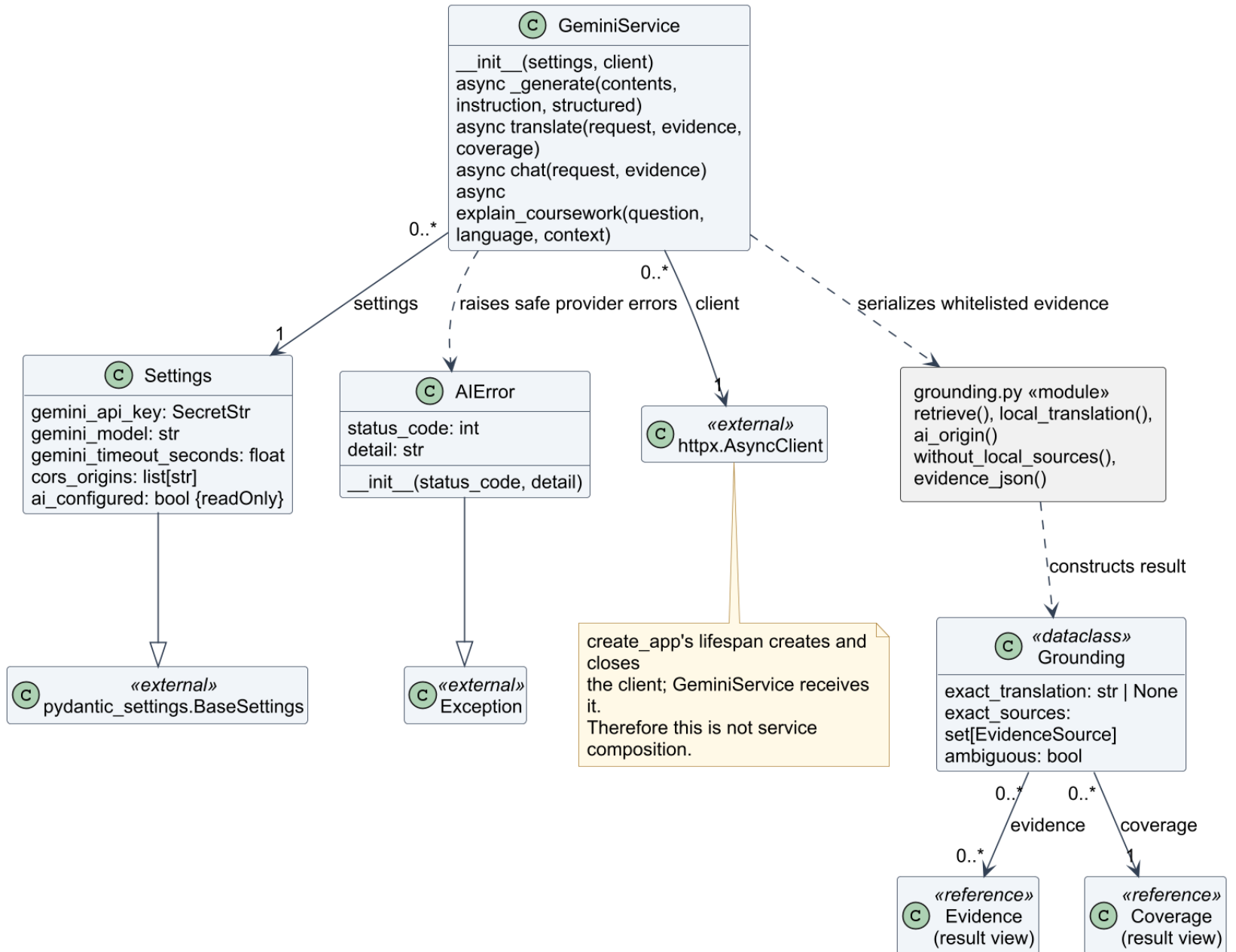
SuggestRequest and all coursework classes inherit RequestModel (directly, except ParseRequest and ExplainRequest as drawn). Other import classes directly inherit BaseModel. SuggestedEntry additionally forbids extra fields; approval cannot be imported. Three member strings may initially be blank; nonblank names must be distinct. UI type aliases/interfaces are not additional runtime model classes.

Gemini wire-response validation classes



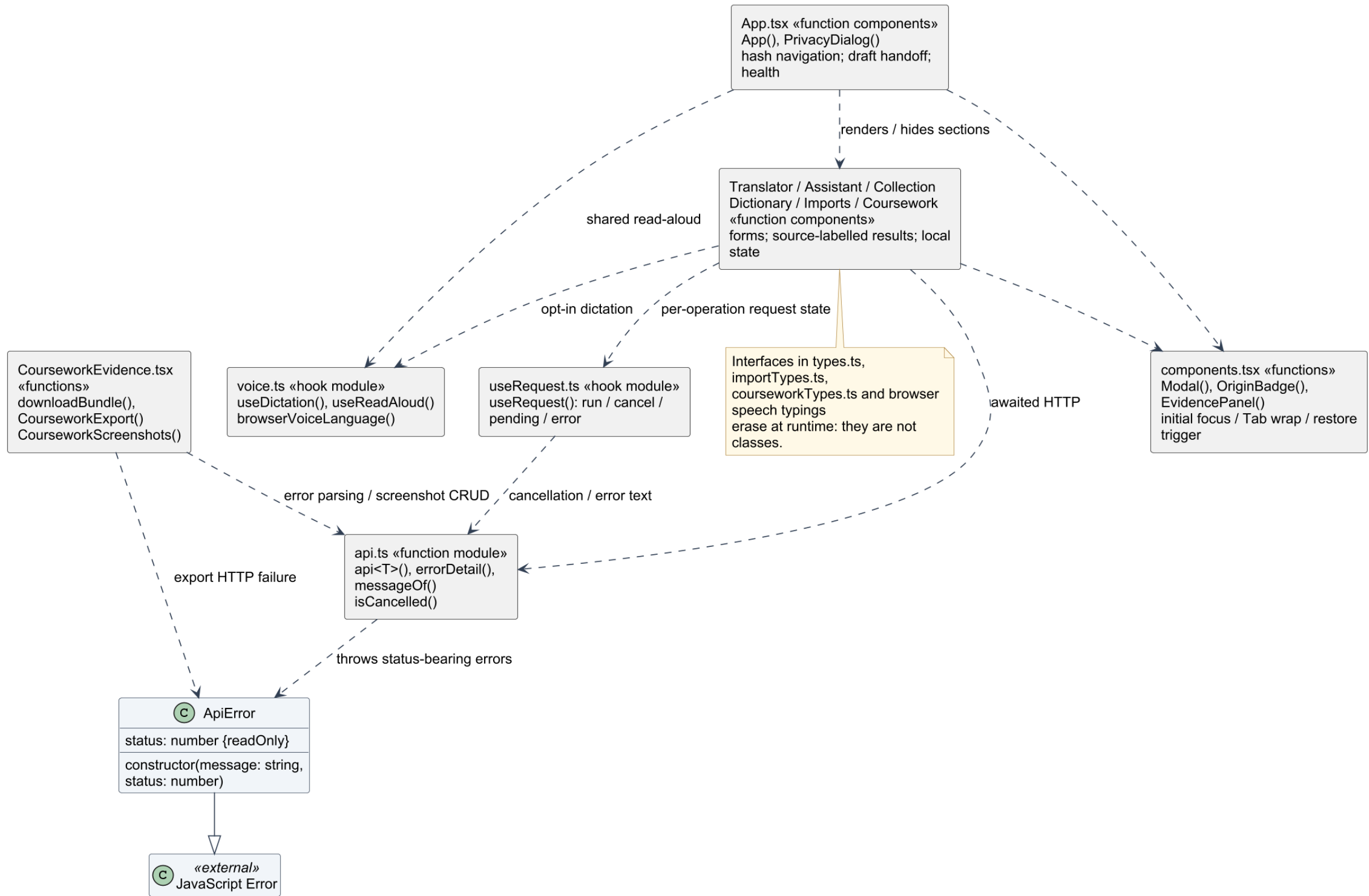
All five primary classes live in `backend.gemini` and extend `BaseModel`.
Optional fields/list defaults justify 0..1 and 0..*.
They validate upstream wire data, not stored corpus entities.
These classes are distinct from public `TranslationContent/ChatResponse` DTOs.

Actual provider service and local grounding result



GeminiService is a real class. Retrieval and route handlers are functions. Grounding is a value holder, not a hidden training or translation service. At most 8 selected evidence records are emitted by `retrieve()`.

Frontend class and function-component structure

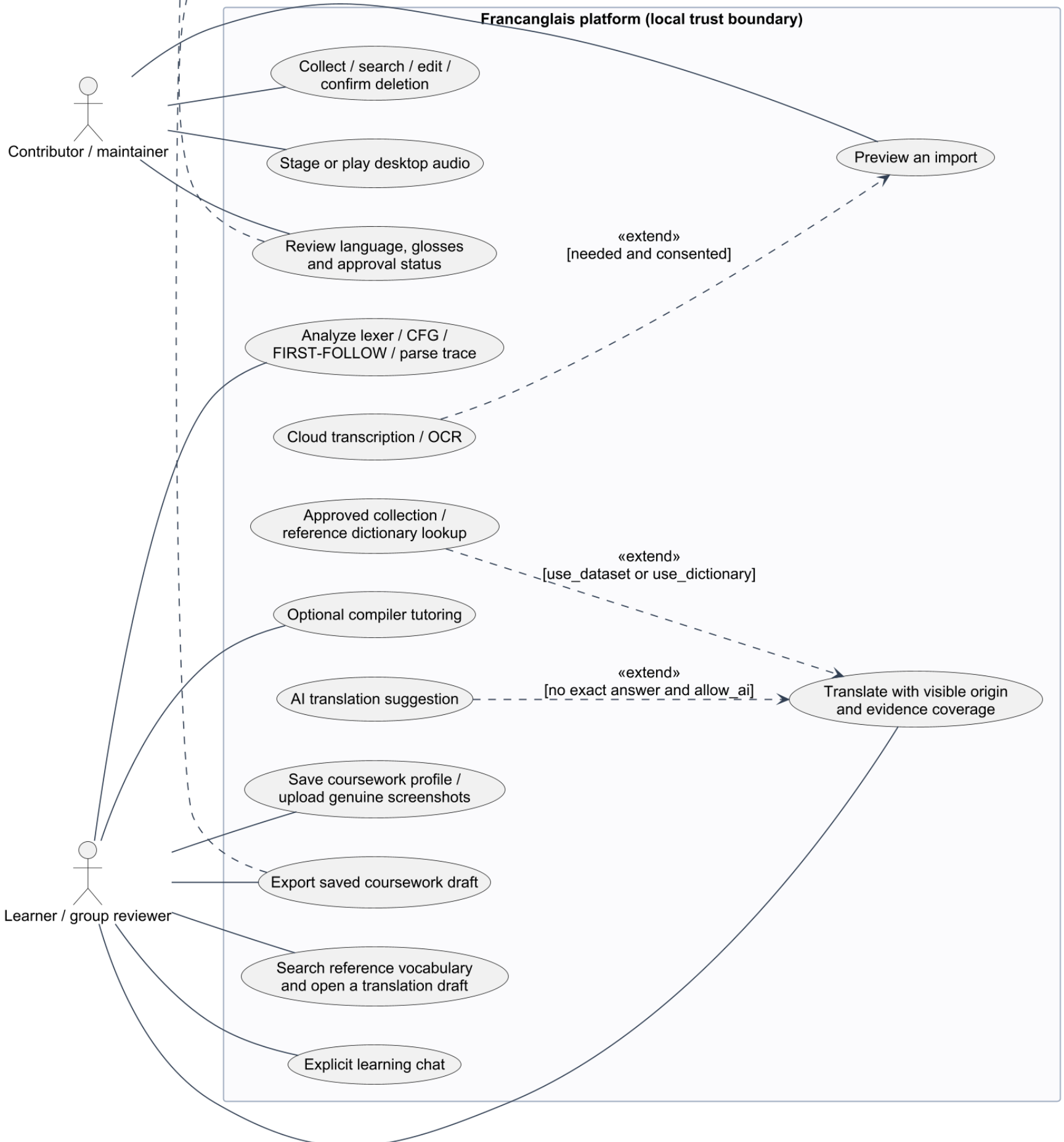


ApiError is the only authored production TypeScript class.
React function components and hooks must not be recast as service classes.
Module dependencies have no association multiplicities.

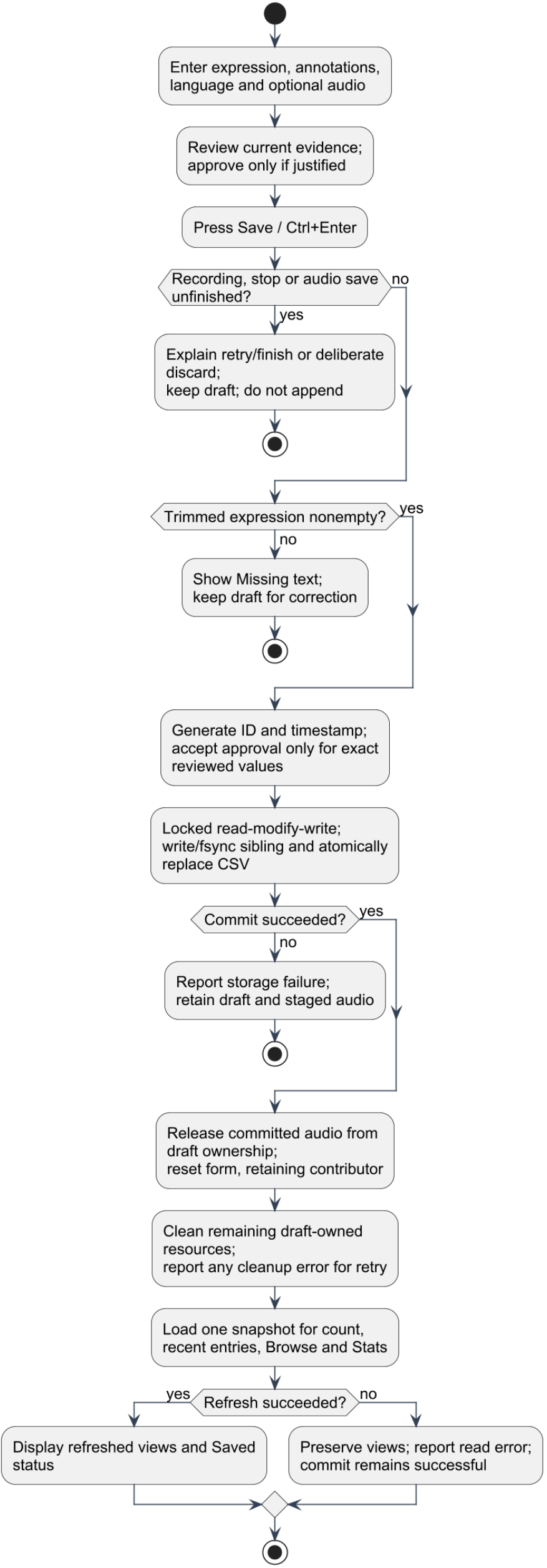
Implemented user goals and explicit optional processing

Draft generation is not completed fieldwork, final report pagination or group sign-off.

Saved does not mean approved. Preview/handoff never auto-saves. Roles are not authentication permissions. Dictionary search/handoff does not create fieldwork.

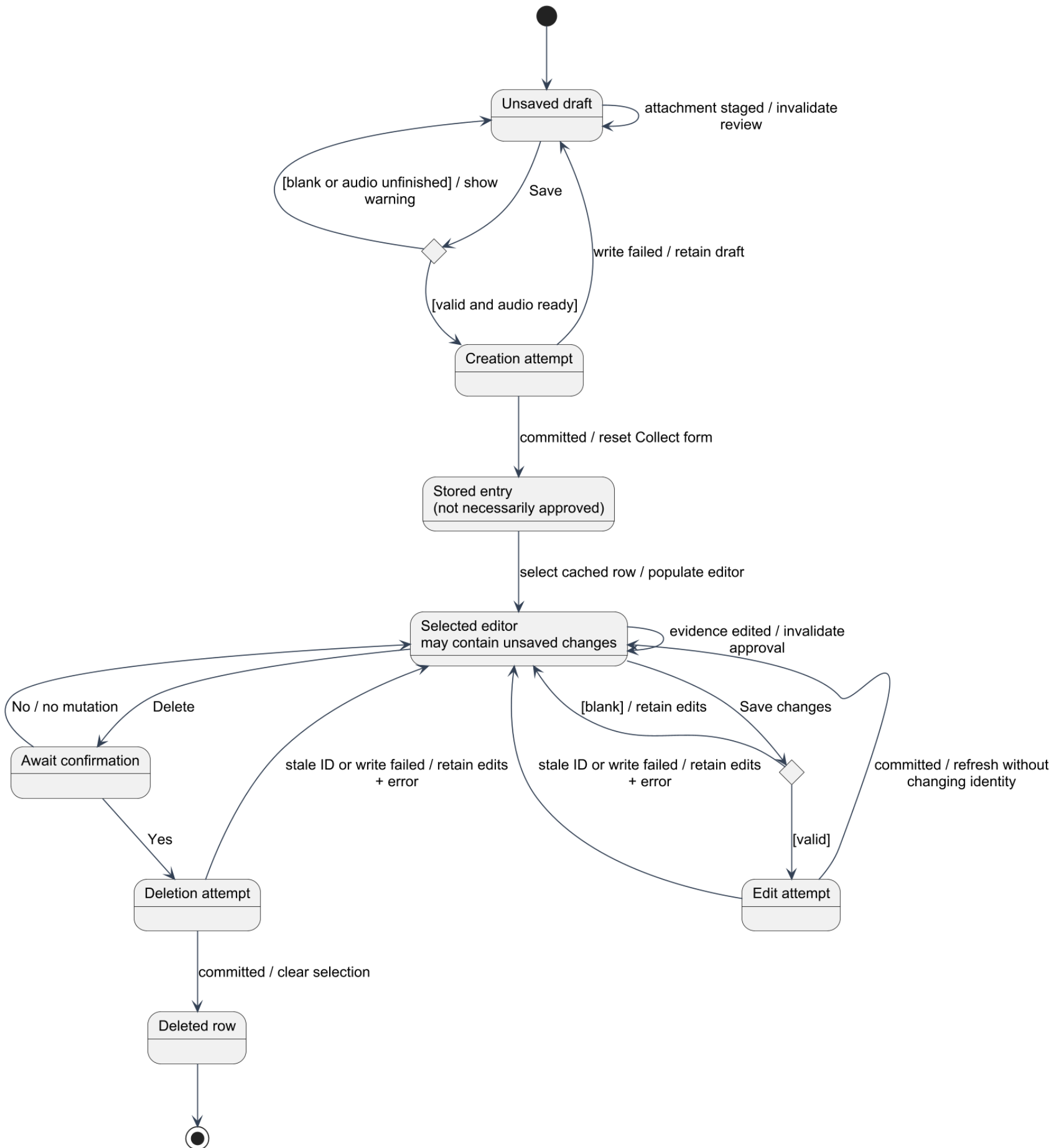


One desktop save attempt, including failure boundaries



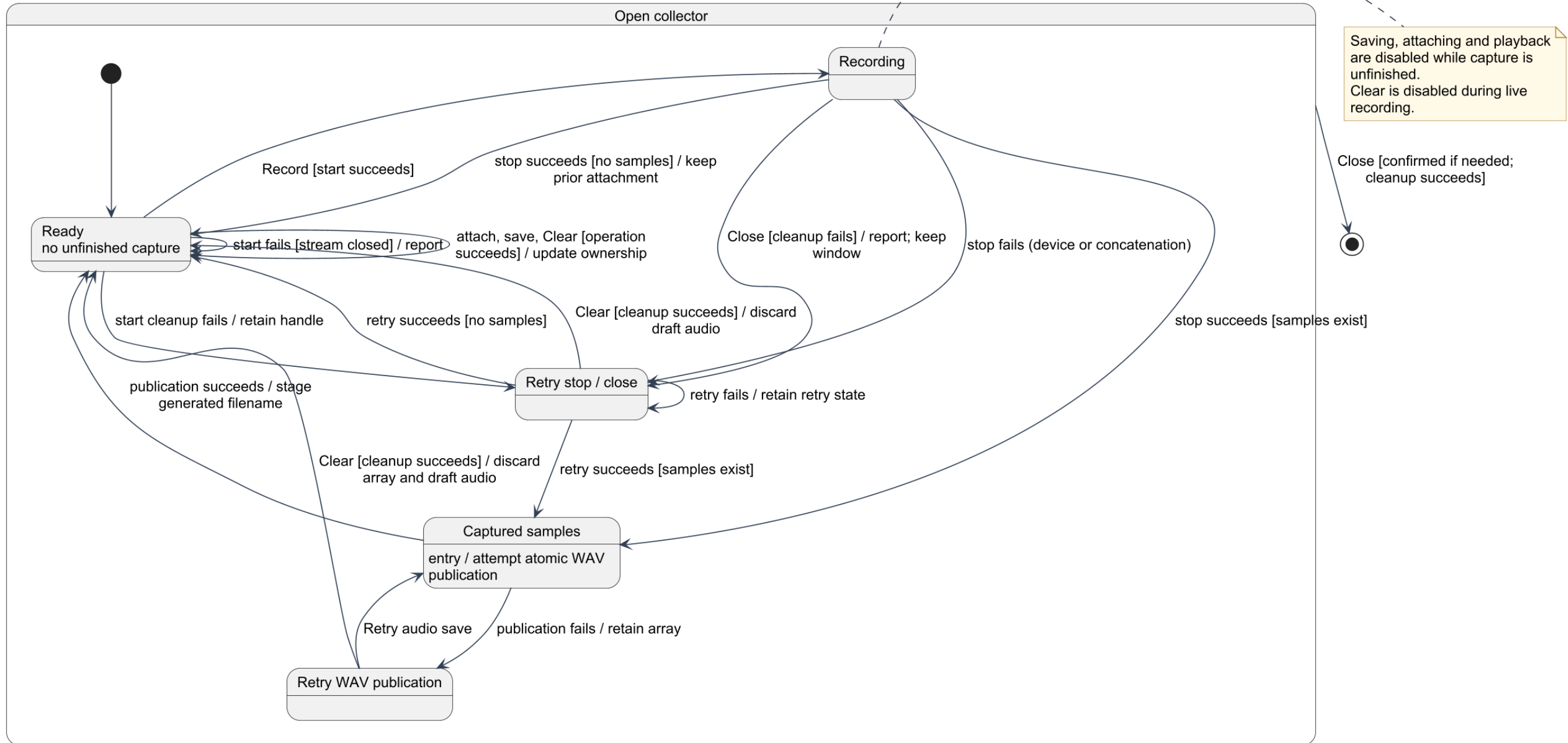
Diamonds represent exclusive guarded decisions, not parallel work.
One final node ends this attempt, not the collector application.
Duplicate warnings are advisory on desktop; API duplicates return 409.
File staging is separate; recording details appear in the sequence view.

Logical entry and selected-editor lifecycle



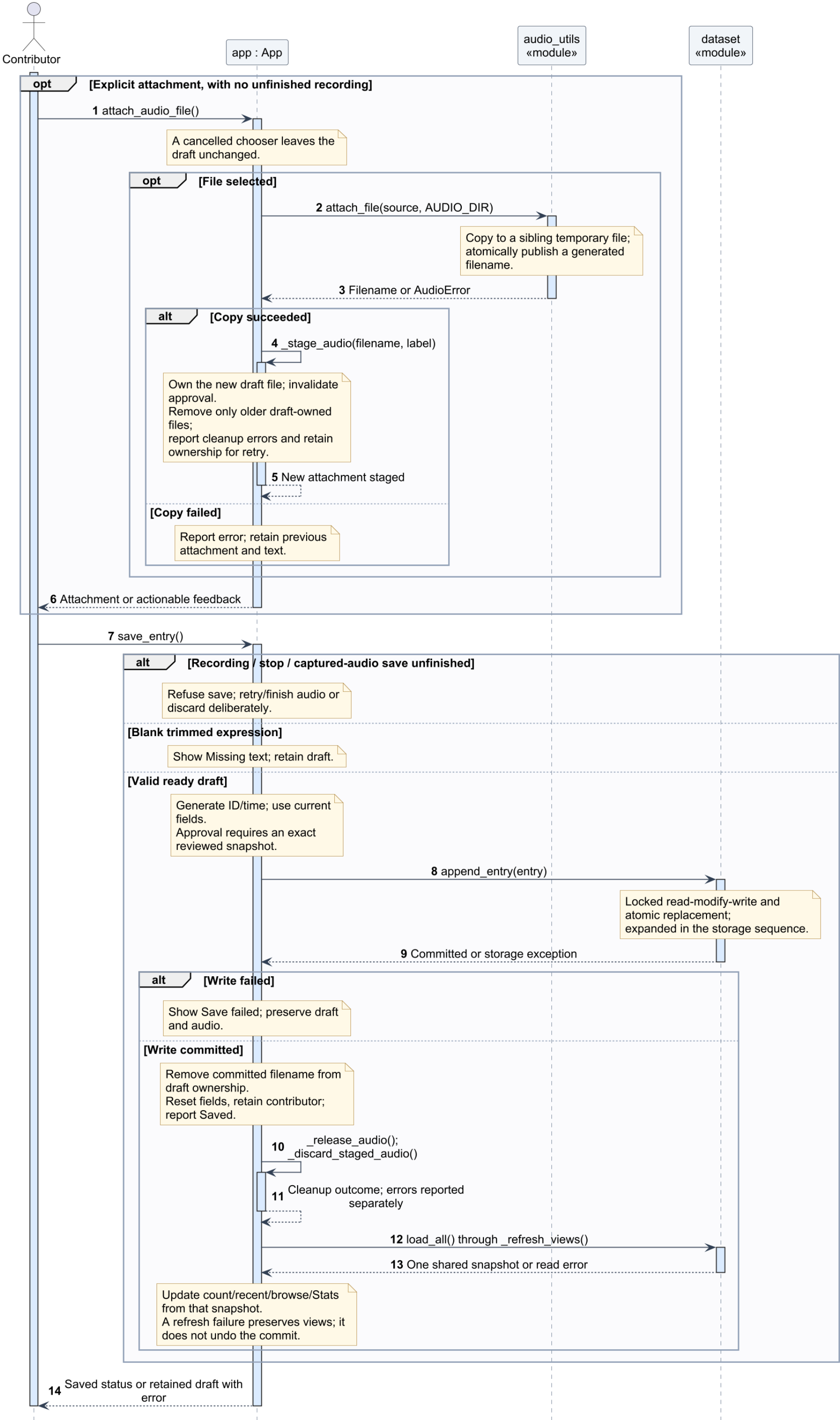
This is a logical row/editor model, not an implemented enum or whole-app lifecycle. The selected editor may outlive a row removed by another writer; stale errors are explicit. A successful commit is not reversed by later display/cleanup failure. Audio staging has a separate lifetime; deleting a saved row retains its recording.

Desktop capture readiness and retry states

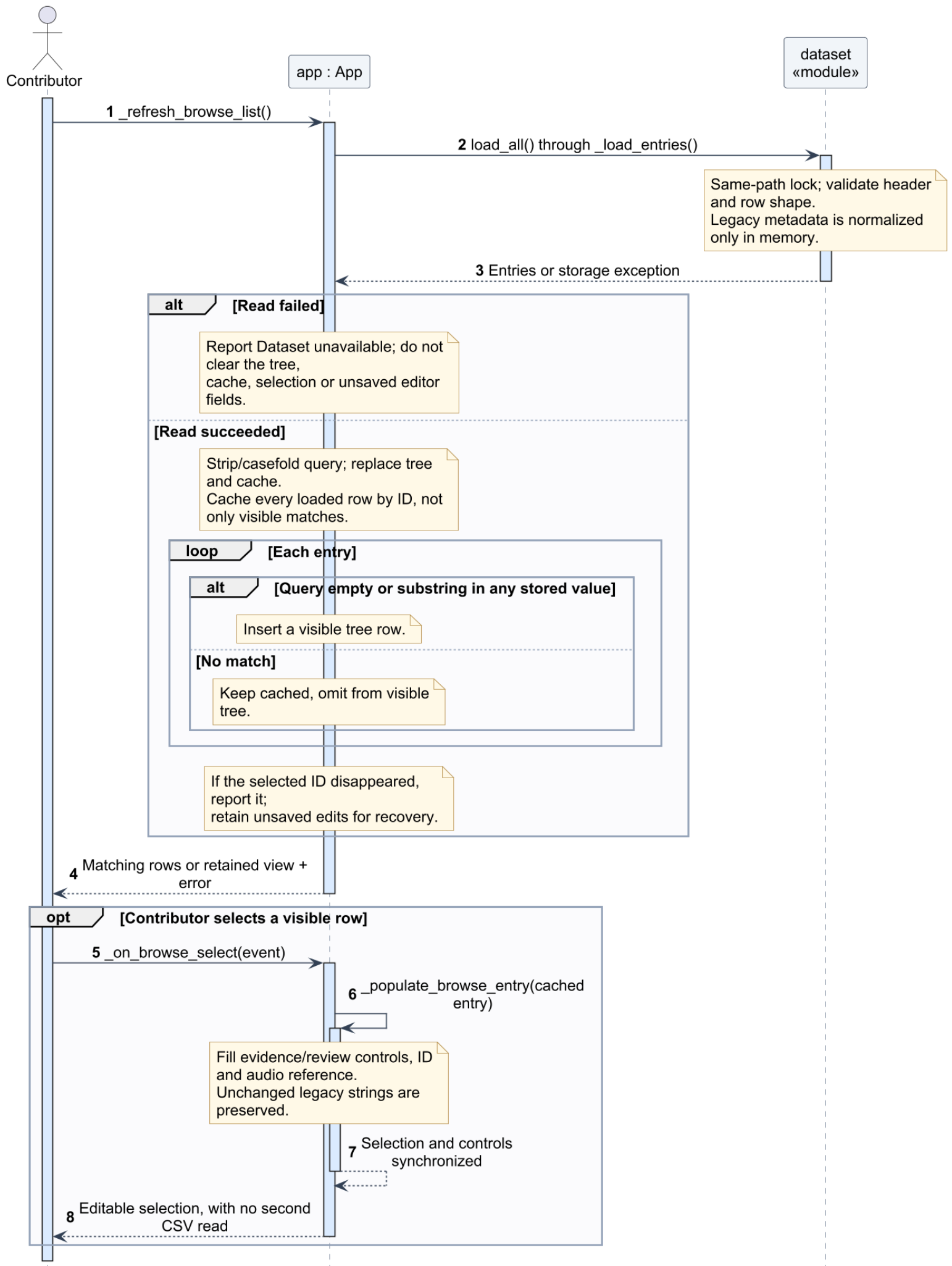


Conceptual states derive from recording, stream, pending-stop and captured-array fields. Ready may contain a staged filename; it does not mean an empty form. Failed cleanup keeps the window/retry state available, not a false closed status. Draft-owned files may be discarded; committed recordings are never owned by the draft.

Desktop attachment and truthful entry saving

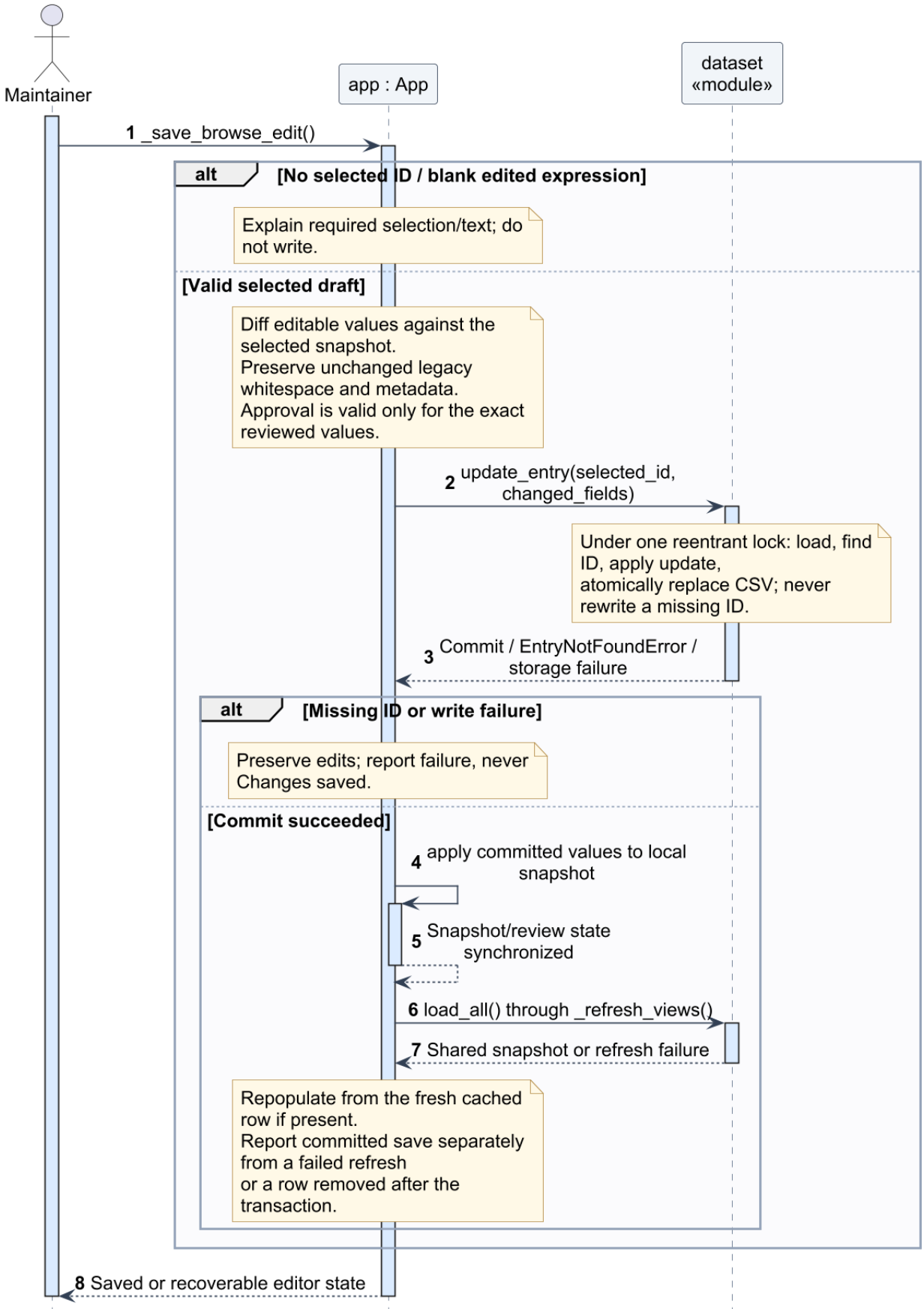


Desktop search, preserved views and cached selection



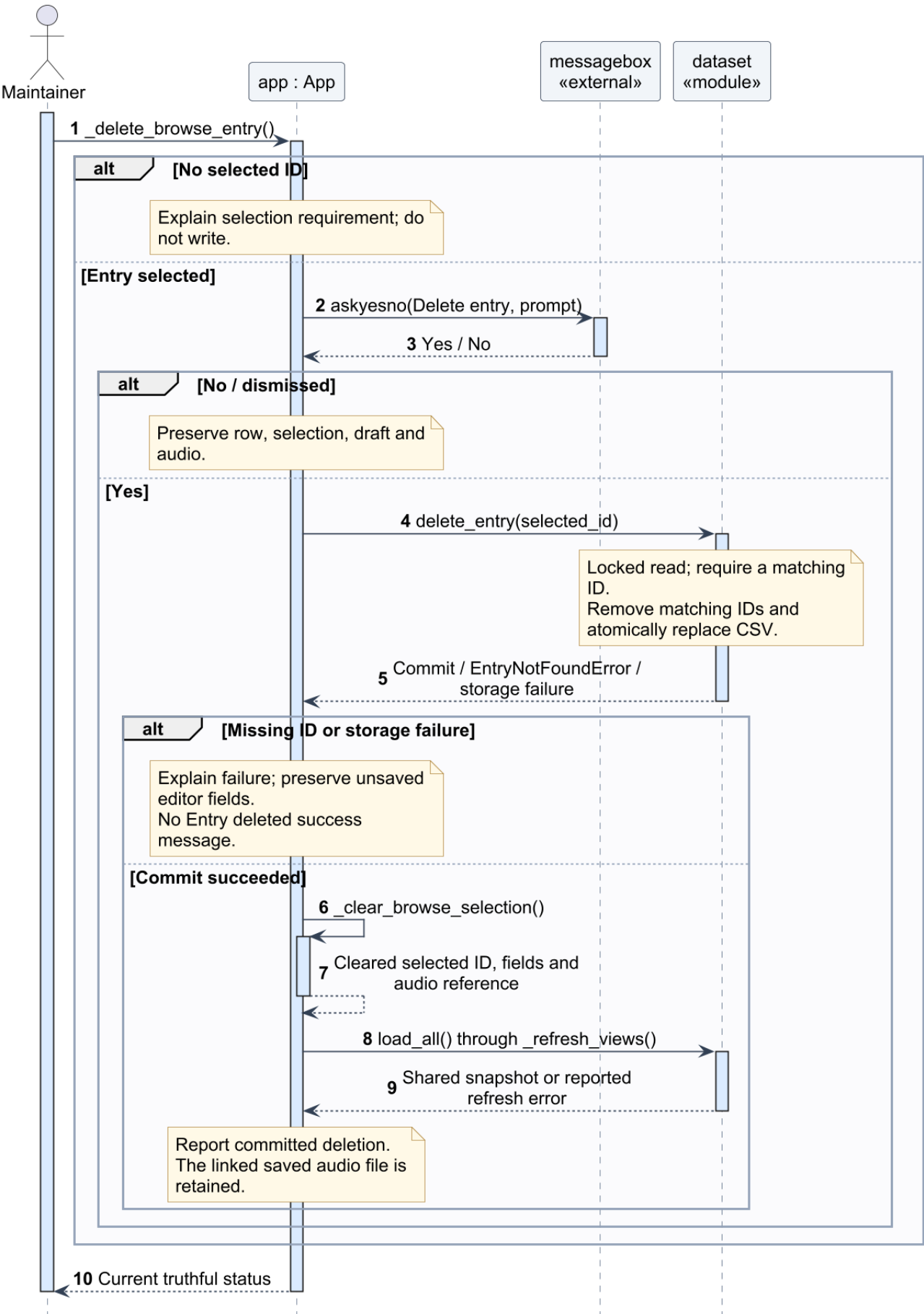
IDs should remain unique; manually corrupted identifiers are not repaired here. Filtering is read-only. Empty results are distinct from a failed read.

Desktop changed-field editing and review integrity



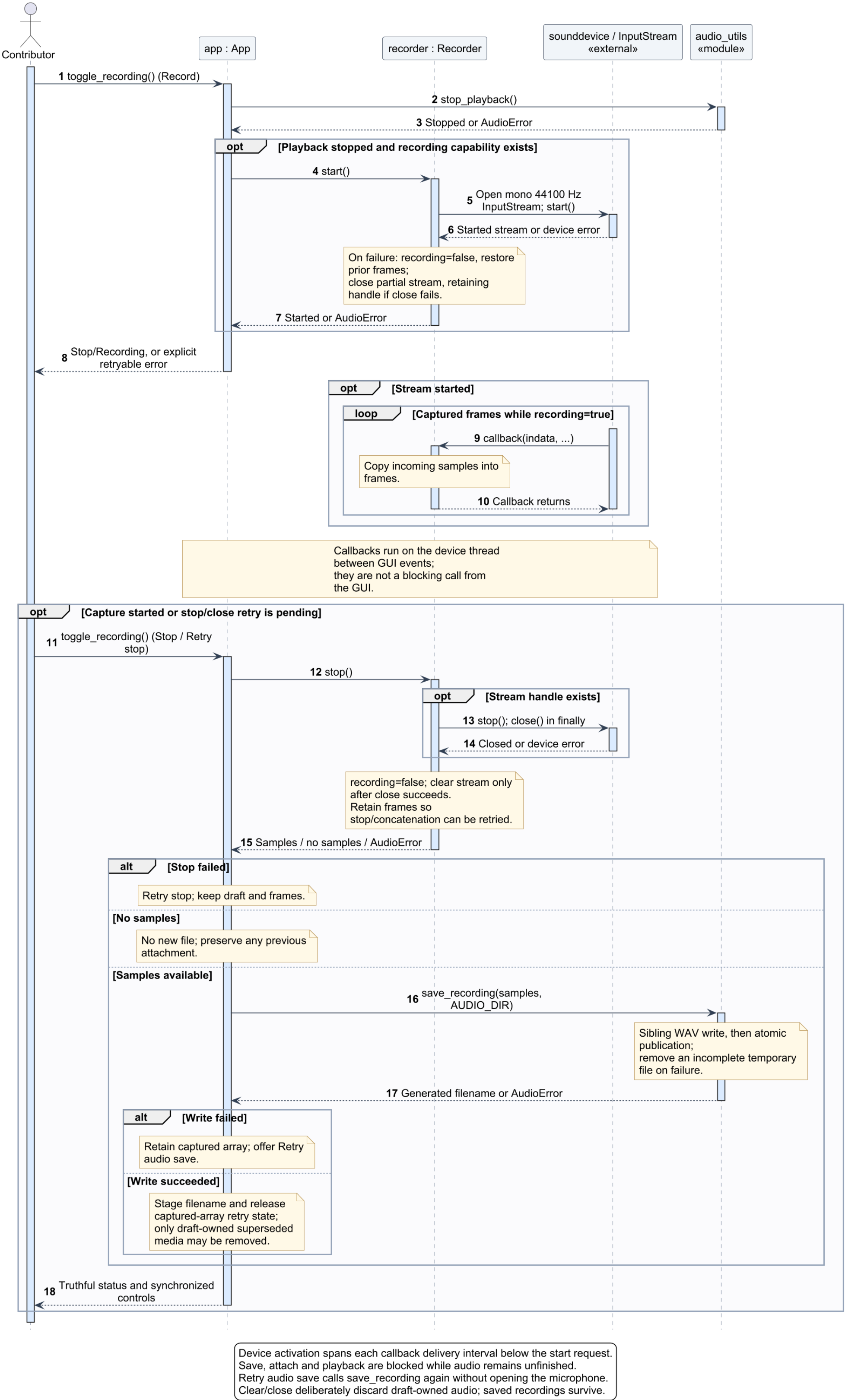
Desktop edits preserve ID, timestamp, contributor and audio reference. Text, glosses, topic, location, notes, language and lexical metadata are editable. Locks prevent interleaved writes, not semantic last-writer-wins conflicts.

Confirmed deletion without false success or media loss

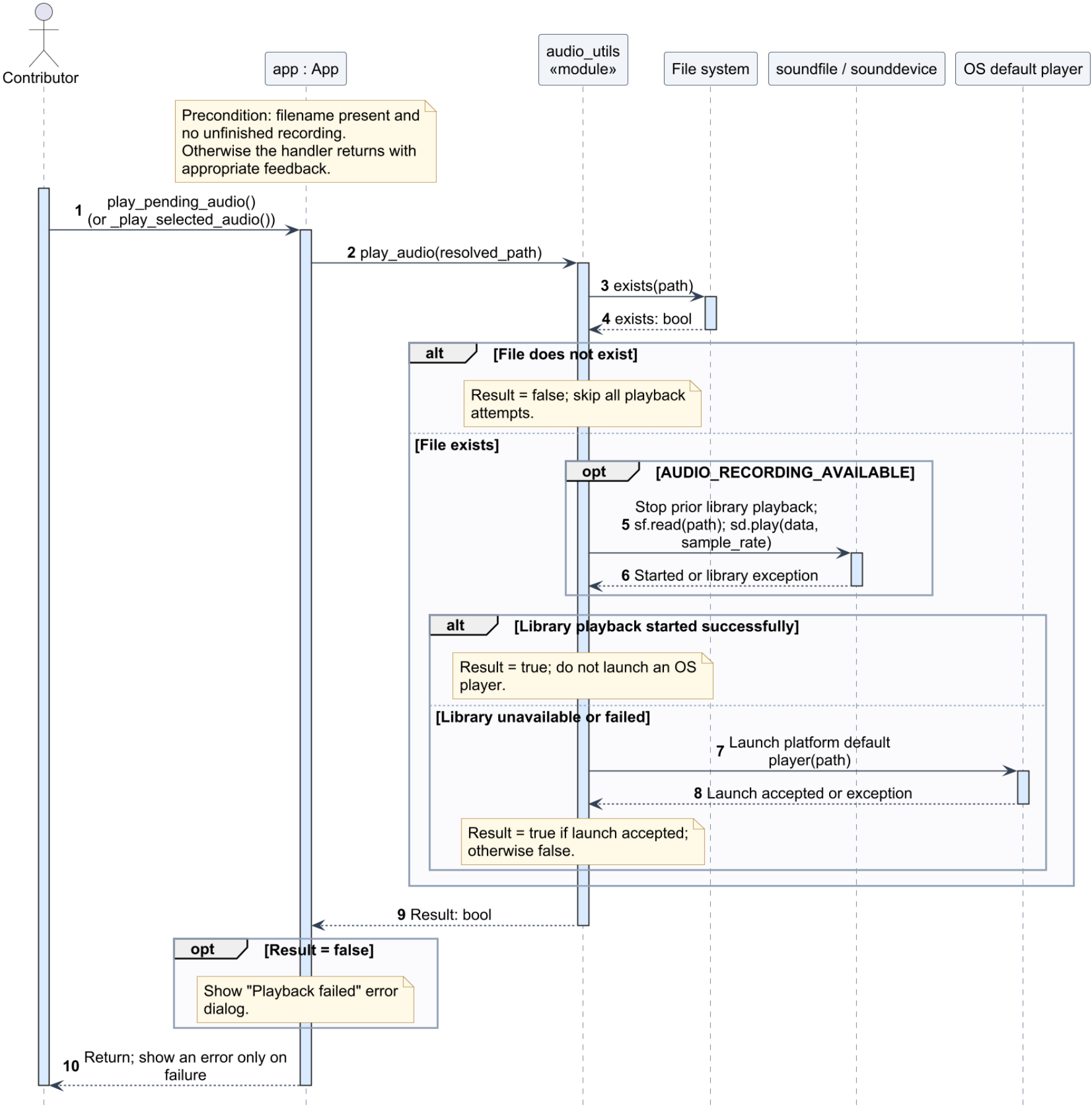


Confirmation is a decision, not an authorization or authentication check.
Atomic replacement protects write integrity; there is no undo stack.

Retryable microphone capture and WAV staging

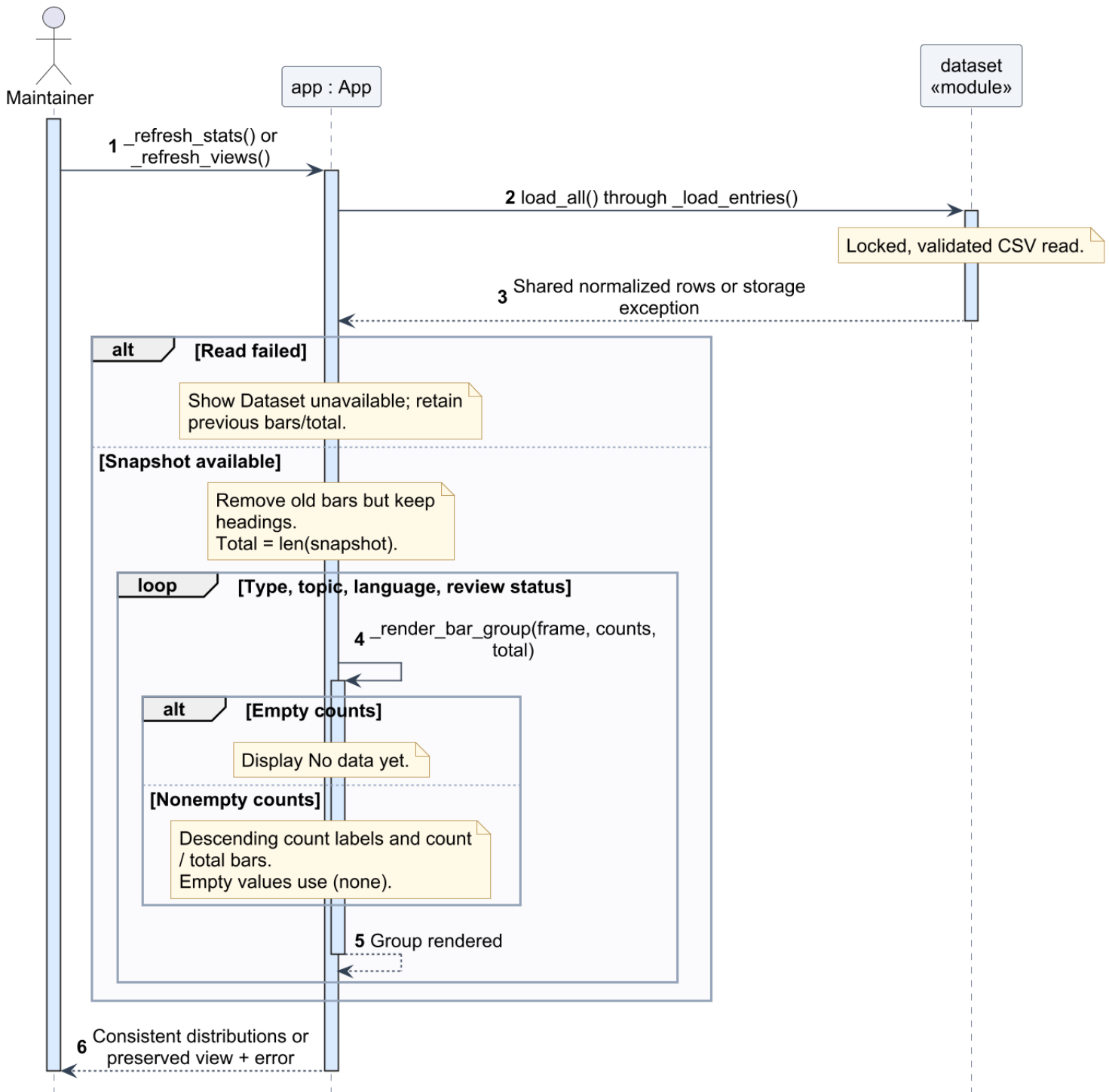


Play Audio with Library and OS Fallback



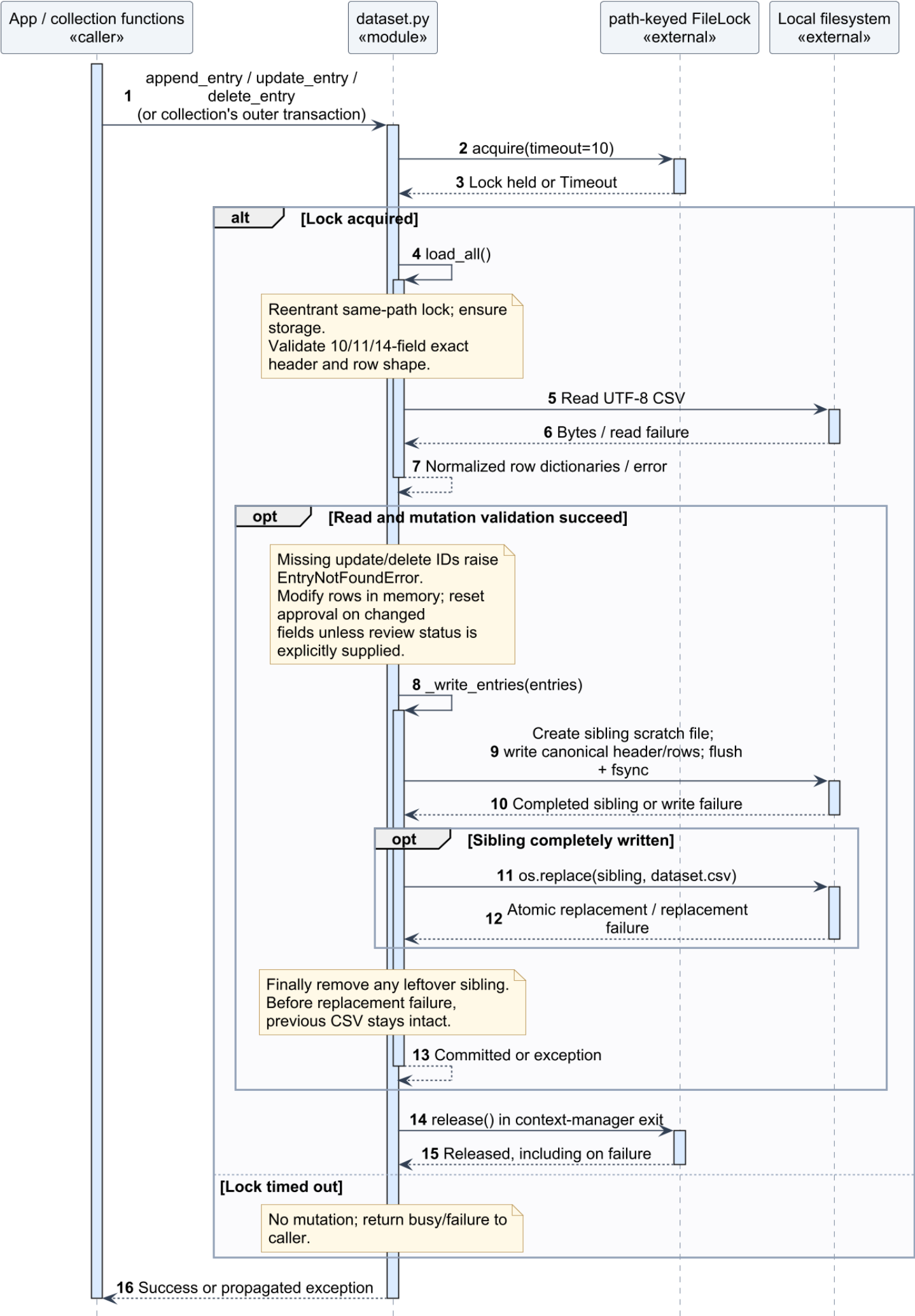
true means a playback request was accepted, not that sound was heard to completion. OS launch acceptance is not playback completion; media decoding is external. Expected audio errors are reported by the GUI; CSV is unchanged. Clear/close can stop in-process playback, not an external OS player's lifetime.

Consistent four-way statistics from one snapshot



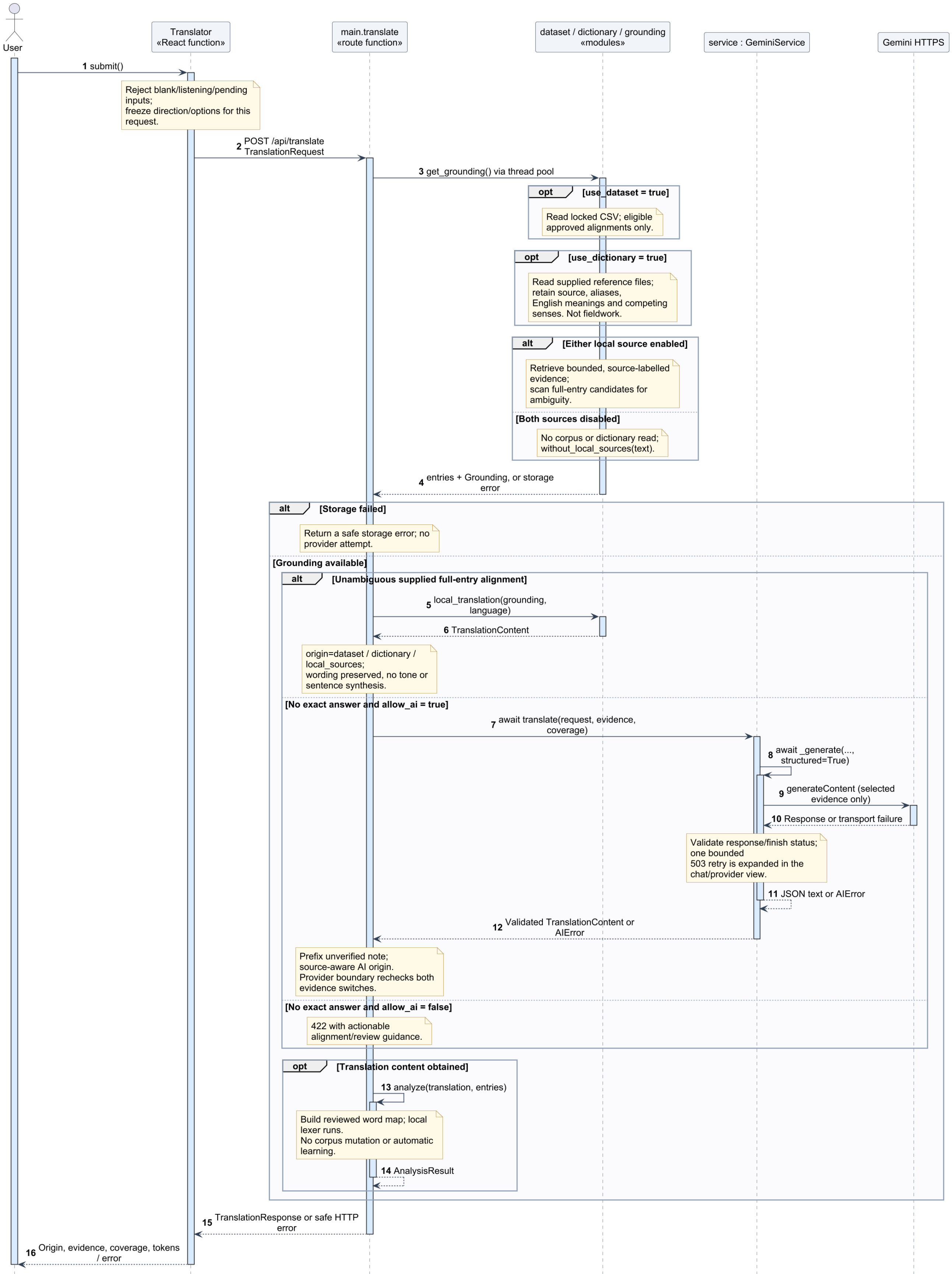
`_refresh_views` also supplies the same rows to count, recent and Browse views. Direct Stats refresh loads once; `count_by()` is not called by this handler. Performance evidence includes active-tab Tk idle rendering, not only CSV timing.

Shared locked CSV mutation and failure-safe replacement



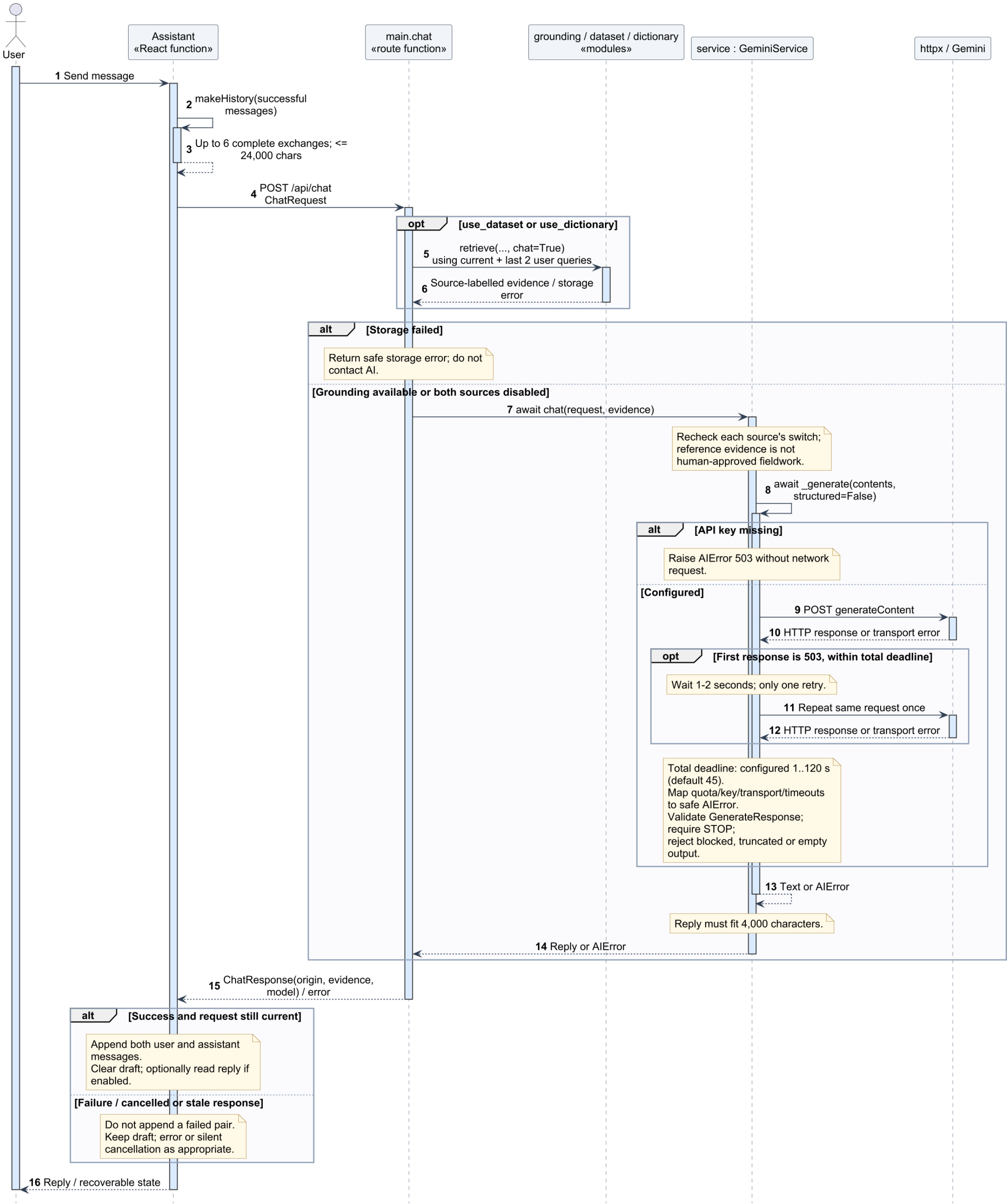
collection functions take the same outer lock around duplicate/ID checks and write. Lock activation shows the acquire/release call, not the duration of file ownership. Append also rewrites the full dataset; there is no unsafe direct append path. Protection covers cooperative users of this path, not external manual writers.

Web translation: enabled local sources, then optional AI



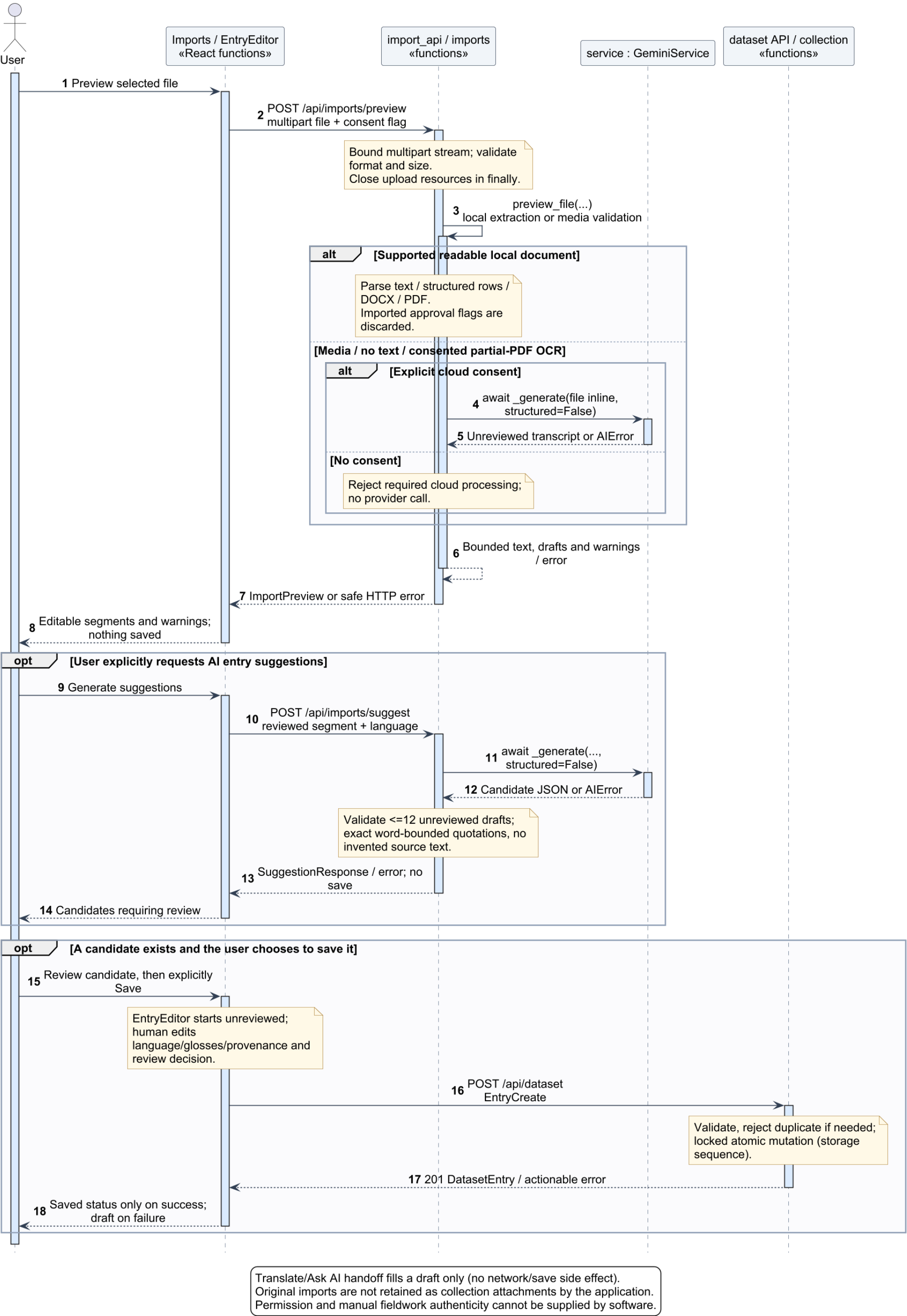
Awaited work is shown as a request/reply interval, not a blocked UI thread. Storage/validation/provider failure exits to an error; it never becomes a success. Cancellation prevents stale display, not guaranteed server/provider rollback.

Explicit chat and bounded provider response handling

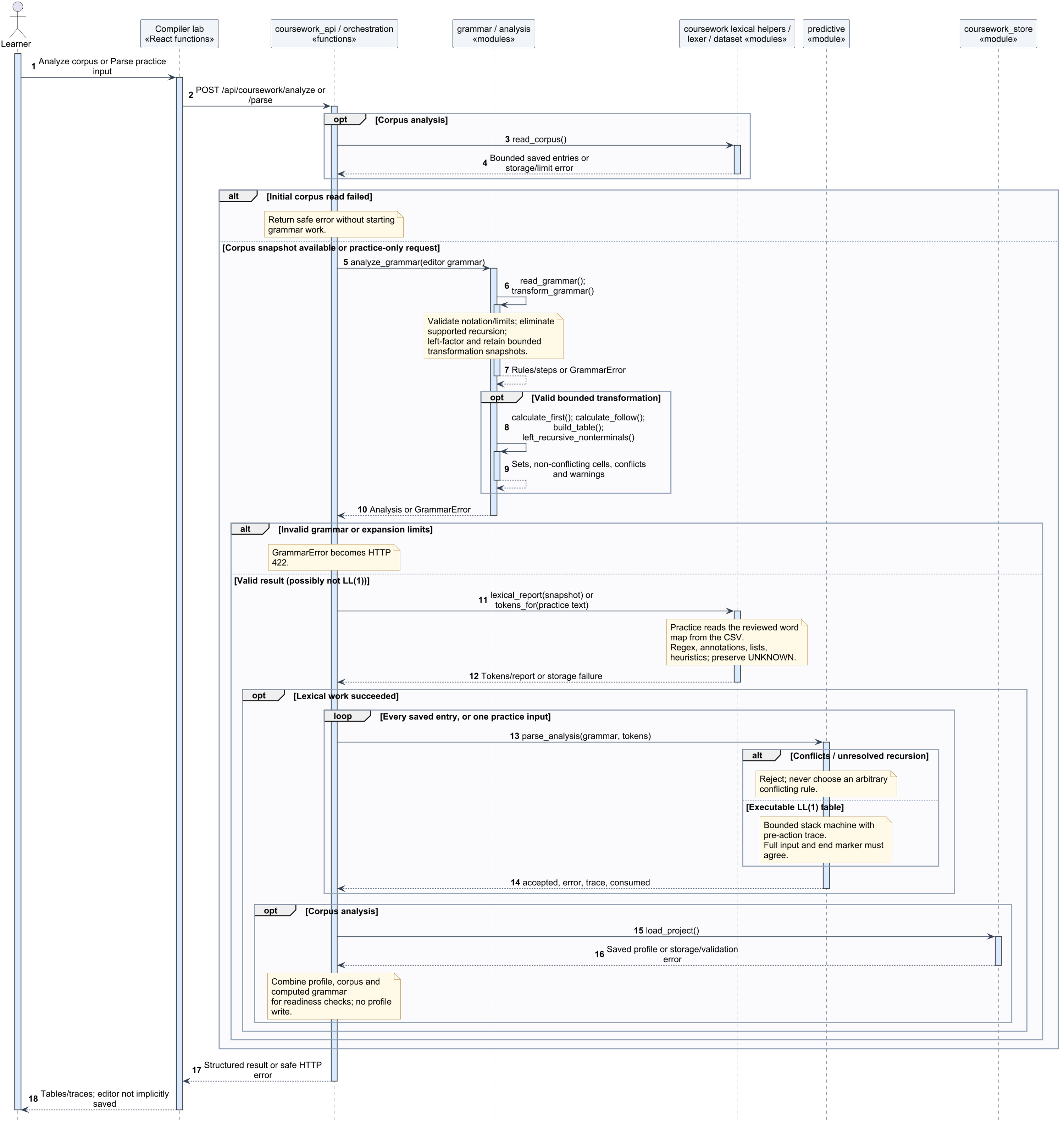


The collection and dictionary switches are independent.
Disabled sources supply no retrieved examples; typed text/history is still sent.
The model never receives contributor/location/notes through retrieval.
Clear removes browser-tab history, not possible provider-retained records.

Import preview, explicit suggestions and human review/save

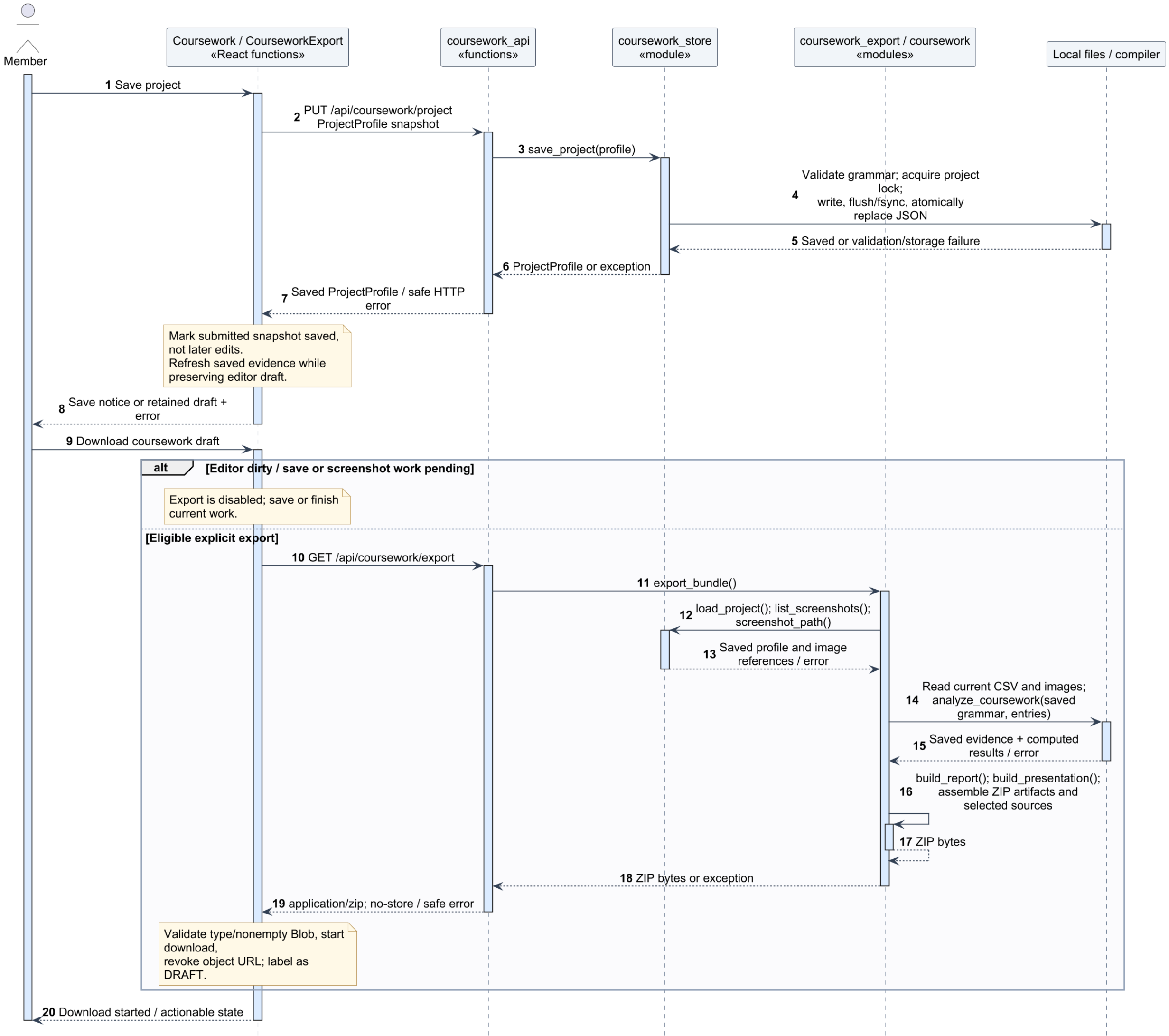


Deterministic grammar analysis and category-token parsing



Module self-calls condense internal function boundaries.
No provider determines parsing. Corpus/grammar fit is not linguistic authenticity.
Existing corpus/media stay unchanged; absent storage can bootstrap an empty header.

Save coursework, then export only saved evidence



No unsaved practice input or AI answer is included.
ZIP includes actual corpus/provenance/profile: review before sharing.
Individual files are protected, but export is not a global multi-file transaction.
Missing research remains visible; 25 sections do not guarantee 25-30 final pages.

Reference lookup: separate sources, explicit draft handoff

